Neural embedding of research papers reveals the research activities of policy research institutes

Munjung Kim[1], Jisung Yoon[2], Hyunuk Kim[3], Woo-Sung Jung[1,2]

[1]Department of Physics, Pohang University of Science and Technology; [2]Department of Industrial and Management Engineering, Pohang University of Science and Technology; [3]Questrom School of Business, Boston University

The policy research institute plays an important role in shaping national agendas by providing evidence to policymakers. Many countries have recently introduced a national evaluation on their research institutes, fulfilling the requirements for the optimal allocation of resources. However, for policy reports, the lack of qualitative measures hinders fair evaluations. For this purpose, several quantitative methods (e.g., topic modeling) have been proposed to overcome this issue, but they still have some limitations in that they could not characterize institutes' research activities regarding diverse and rapidly changing topics. Here, we present a neural embedding approach that learns a vector representation of research papers as a new solution to the aforementioned issues and elucidate that this approach can determine the characteristics of each institute [1].

We source 21,397 policy research papers published from 2000 to 2018 by 27 national policy research institutes in the Republic of Korea. Each research paper is represented as a single vector by doc2vec method (Fig. 1a). We visualize the embedding space by projecting embedding vectors into a two-dimensional space and observe that the same or similar departments are well clustered (Fig. 1b). In the embedding space, the mean vector of each institute's papers can be considered the overall interest of the institute. Based on this concept, we define the *diversity* of the topic in each institute's papers as the average cosine distance from the embedding vectors of the papers to the overall interest (the mean vector of each institute). In addition, the *temporal change of the diversity* is regarded as how much the research areas of the institute have expanded or shrunk. We introduce *topical change*, defined as the cosine distance between the mean vectors in different years, to quantify how much overall research interests have shifted across periods.

According to the results, the Korean Women Development Institute (KWDI) is classified as the most generalized and expanding institute between 2000 and 2018. This finding is consistent with the emergence of women's issues over the past two decades in Korea [2]. In addition, the Korean Development Institute (KDI) and Korea Information Society Development Institute (KISDI) have quickly changed their overall topics (Fig. 1e), which is compatible with the government's appraisal report. Therefore, we conclude that our approach can help to easily grasp the characteristics of policy research institutes both numerically and systemically. This capability to capture the characteristics of the institutes can serve as a new quantitative analytic frame for policy research.

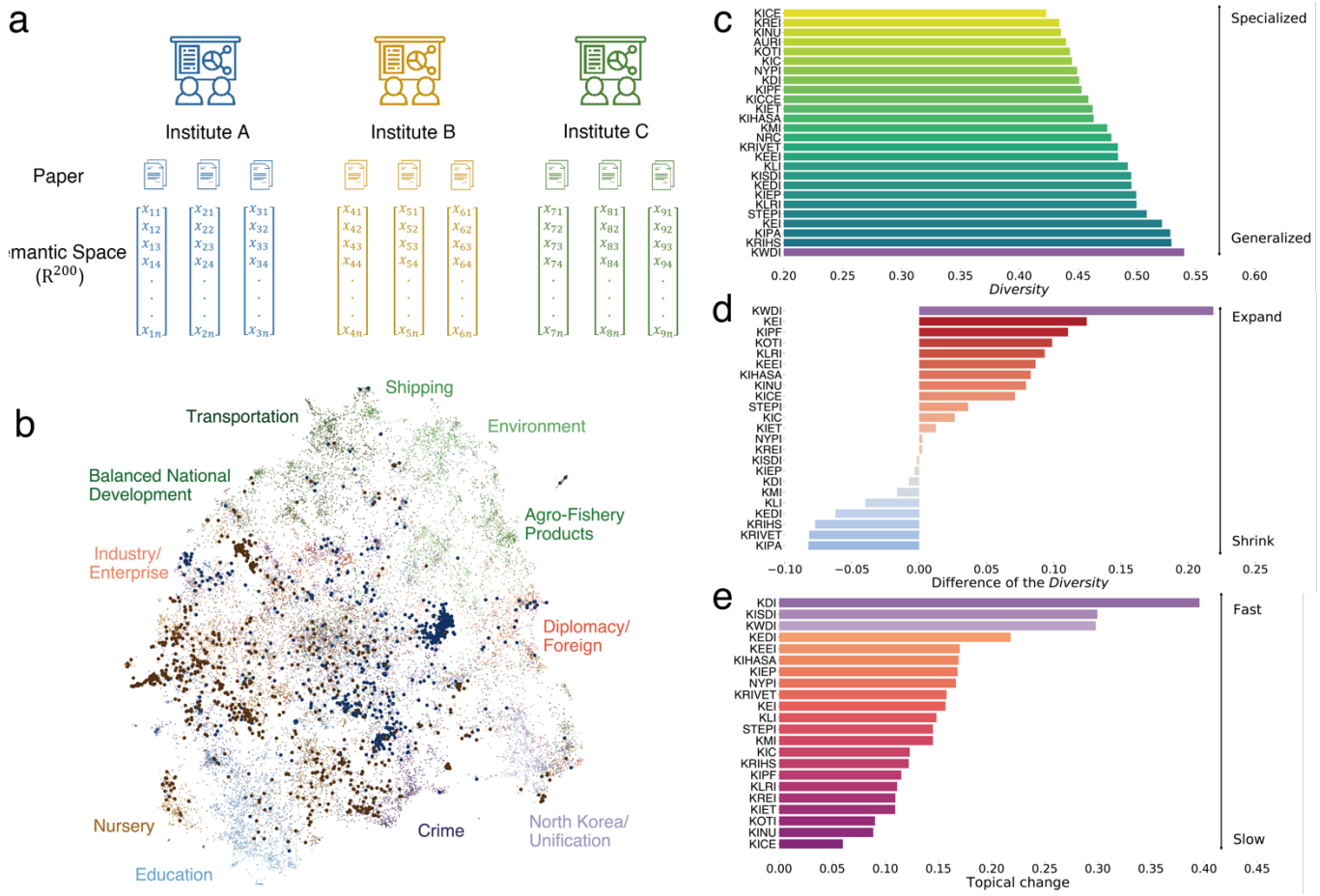

Figure 1: Neural embedding vectors of research papers effectively elucidate the characteristics of research institutes. a. Schematics diagram. **b.** Topological map of embedding vectors projected with applying a dimension reduction method on the obtained vectors. Each dot represents a paper, and the color of the dot indicates its affiliation. Larger dots represent KWDI (brown) and KISDI (blue). **c.** Overall diversity of institutes. **d.** Difference in the diversity between 2000 and 2018. **e.** Topical change between 2000 and 2018.

[1] Le, Q., & Mikolov, T. Distributed representations of sentences and documents. ICML, pp. 1188-1196 (2014).
[2] Research institute evaluation reports, National Research Council, https://www.nrc.re.kr/board.es?mid=a10301000000&bid=0008 (in Korean)